

Description of Course CSE 402

PART A: General Information

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|----------------------------------|--|
| 1 Course Title | : Numerical Analysis, Simulation and Modeling
Sessional |
| 2 Type of Course | : Sessional |
| 3 Offered to | : Department of CSE |
| 4 Pre-requisite Course(s) | : None |

PART B: Course Details

1. Course Content (approved by the Academic Council)

Sessional based on CSE 401(Numerical Analysis, Simulation and Modeling)

2. Course Objectives

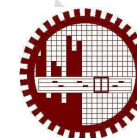
The students are expected to:

- Implement numerical methods for root-finding, systems of equations, and optimization.
- Apply curve fitting, interpolation, and numerical integration to engineering problems.
- Develop and validate simulation models using Monte Carlo and discrete event methodologies.

3. Knowledge required

Technical

- Computer Programming
- Mathematics (Calculus and Linear Algebra)



Analytical

- Algorithm Design
- Error Analysis

4. Course Outcomes (COs)

CO No.	CO Statement	Corresponding PO(s)	Domains and Taxonomy level(s)	Delivery Method(s) and Activity(-ies)	Assessment Tool(s)
CO 1	Implement and analyze numerical algorithms for solving non-linear equations and linear systems.	PO1, PO2, PO4	C3, C4	Lab session and demonstration	Assignment, Online, Quiz
CO 2	Apply interpolation, regression, and numerical integration techniques to model data and approximate functions.	PO1, PO3	C3	Lab session and demonstration	Assignment, Online, Quiz
CO 3	Design and evaluate simulation models using random number generation and Monte Carlo methods.	PO1, PO3, PO5	C3, C5	Lab session, demonstration, and project-oriented tasks	Assignment, Online, Quiz

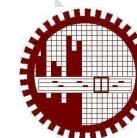
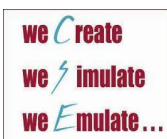
*Program Outcomes (POs)

PO1: Engineering knowledge; PO2: Problem analysis; PO3: Design/development of solutions; PO4: Investigation; PO5: Modern tool usage; PO6: The engineer and society; PO7: Environment and sustainability; PO8: Ethics; PO9: Individual work and teamwork; PO10: Communication; PO11: Project management and finance; PO12: Life-long learning.

**Domains

C-Cognitive: C1: Knowledge; C2: Comprehension; C3: Application; C4: Analysis; C5: Synthesis; C6: Evaluation

A-Affective: A1: Receiving; A2: Responding; A3: Valuing; A4: Organizing; A5: Characterizing



P-Psychomotor: P1: Perception; P2: Set; P3: Guided Response; P4: Mechanism; P5: Complex Overt Response; P6: Adaptation; P7: Organization

5. Mapping of Knowledge Profile, Complex Engineering Problem Solving and Complex Engineering Activities

COs	K1	K2	K3	K4	K5	K6	K7	K8	P1	P2	P3	P4	P5	P6	P7	A1	A2	A3	A4	A5
CO1		√							√		√									
CO2		√							√		√									
CO3		√							√		√							√		

K-Knowledge Profile:

K1: A systematic, theory-based understanding of the natural sciences applicable to the discipline; **K2:** Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline; **K3:** A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline; **K4:** Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline; **K5:** Knowledge that supports engineering design in a practice area; **K6:** Knowledge of engineering practice (technology) in the practice areas in the engineering discipline;

K7: Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, environmental and sustainability; **K8:** Engagement with selected knowledge in the research literature of the discipline

P-Range of Complex Engineering Problem Solving:

P1: Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach; **P2:** Involve wide-ranging or conflicting technical, engineering and other



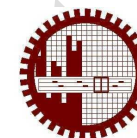
issues; **P3**: Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models; **P4**: Involve infrequently encountered issues; **P5**: Are outside problems encompassed by standards and codes of practice for professional engineering; **P6**: Involve diverse groups of stakeholders with widely varying needs; **P7**: Are high level problems including many component parts or sub-problems

A-Range of Complex Engineering Activities:

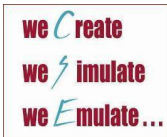
A1: Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information and technologies); **A2**: Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues; **A3**: Involve creative use of engineering principles and research-based knowledge in novel ways; **A4**: Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation; **A5**: Can extend beyond previous experiences by applying principles-based approaches

6. Lecture/ Activity Plan

Week	Topic	Course Outcomes
1	No Lab	-
2	No Lab	-
3	Lab Assignment 1(A1/B1/C1) : Approximations, round-off errors, truncation errors; Visualization and plotting; Root finding: bisection method, false position method, Newton-Raphson method, Bairstow's method (Newton-Raphson/Bairstow) including all relevant topics covered in theory class	CO1
4	Lab Assignment 1(A2/B2/C2) : Approximations, round-off errors, truncation errors; Visualization and plotting; Root finding: bisection method, false position method, Newton-Raphson method, Bairstow's method (Newton-Raphson/Bairstow) including all relevant topics covered in theory class	CO1



Week	Topic	Course Outcomes
5	Lab Assignment 2(A1/B1/C1): Solution of systems of equations: Gauss elimination method, Gauss-Jordan elimination method, LU decomposition; Eigenvalue decomposition: power method including all relevant topics covered in theory class	CO1
6	Lab Assignment 2(A2/B2/C2): Solution of systems of equations: Gauss elimination method, Gauss-Jordan elimination method, LU decomposition; Eigenvalue decomposition: power method including all relevant topics covered in theory class	CO1
7	Reserved Project Group Information Submission: A group will consist of 5 students.	-
8	Project Proposal	CO1, CO2, CO3
9	Lab Assignment 3(A1/B1/C1): Random Number Generation, Monte Carlo Methods (Metropolis-Hastings) including all relevant topics covered in theory class Home Assignment 1: QR method: Optimization: golden-section search, Newton's method, gradient methods, constrained optimization	CO3
10	Lab Assignment 3(A2/B2/C2): Random Number Generation, Monte Carlo Methods (Metropolis-Hastings) including all relevant topics covered in theory class Home Assignment 1: QR method: Optimization: golden-section search, Newton's method, gradient methods, constrained optimization	CO3
11	Lab Assignment 4(A1/B1/C1): Curve fitting, interpolation and approximation: least squares regression, linear interpolation, Lagrange polynomial interpolation, Newton's polynomial interpolation, spline interpolation; Numerical integration and differentiation: Newton-Cotes integration, trapezoidal rule, Simpson's rule, Romberg's integration, Richardson's extrapolation including all relevant topics covered in theory class	CO2
12	Lab Assignment 4(A2/B2/C2): Curve fitting, interpolation and approximation: least squares regression, linear interpolation, Lagrange polynomial interpolation, Newton's polynomial interpolation, spline interpolation; Numerical integration and differentiation: Newton-Cotes integration, trapezoidal rule, Simpson's rule,	CO2



Week	Topic	Course Outcomes
	Romberg's integration, Richardson's extrapolation including all relevant topics covered in theory class	
13	Quiz	CO1, CO2, CO3
14	Final Project Update & Report Submission	CO1, CO2, CO3

7. Assessment Strategy

- **Class Attendance:** Attendance is recorded for every lab session.
- **Lab Assignments (Online/Offline):** Performance is evaluated through regular online coding tasks and offline problem sets.
- **Project Work:** Evaluation includes the initial proposal, periodic updates, and a comprehensive final report.
- **Final Quiz:** A summative assessment covering all theoretical and practical aspects of the course.

8. Distribution of Marks

Offline	10%
Online	30%
Project (Proposal, Final Update, and Report)	30%
Quiz	30%
Total	100%

9. Textbook/ Reference

- *Numerical Methods for Engineers* by Steven C. Chapra and Raymond P. Canale.
- *Discrete-Event System Simulation* by Jerry Banks, John S. Carson II, Barry L. Nelson, and David M. Nicol.